Computer Laboratory B Logging and Reservation System: User Management

Midterm Exam

Object-Oriented Programming 2

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I. Overview

This midterm project is for the user management component of the Computer Laboratory B Logging and Reservation System (CLBLRS) using Spring Boot, Spring Web, mySQL, JDBC Connector, Spring Data JPA, and Thymeleaf.

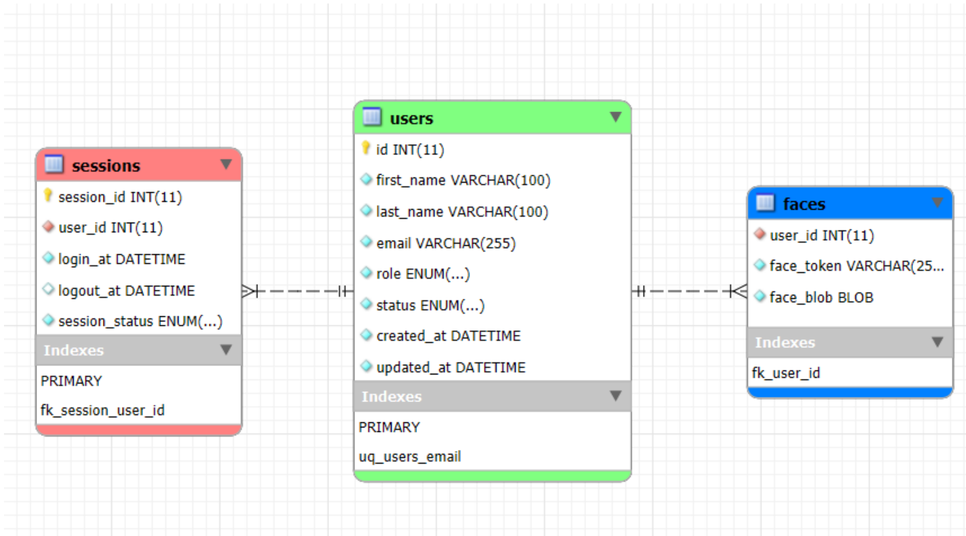
II. Roles

Name	Role
Alvey	Full Stack Developer Front End Lead Developer
Carl	Project Leader Full Stack Developer GitHub Repository Manager
Denzel	Full Stack Developer Documentation Lead
Joeffrey	Database Developer Back End Developer
Kenji	Quality Assurance Front End Developer Documentation Contributor
Meshari	Front End Designer Documentation Contributor
Prof. Paul	Client Project Adviser

III. Functions

The project offers create, read, update, and delete (CRUD) operations for the admin or laboratory staff. These functions are essential for creating new users, getting user information, updating user information, and deleting user records.

IV. Entity Design

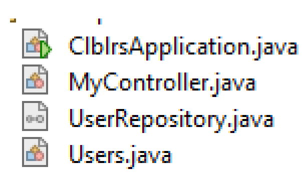


User accounts and privileges for the project made it possible for users to make accounts with certain privileges for specific members of the school. The roles are student, faculty, and admin.

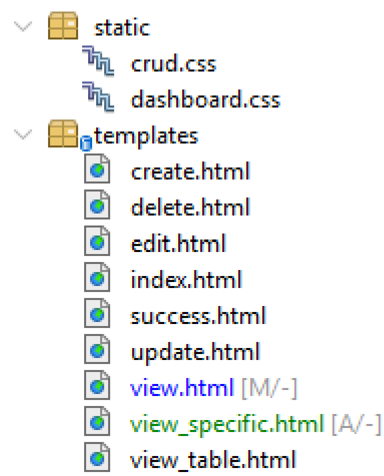
The 'tables' faces and 'sessions' will be used for the user registration and login system and references the primary key 'id' in the table 'users' as foreign key for referential integrity.

V. Architecture

Using model, view, controller (MVC) architecture pattern, the program was constructed with the following classes: controller, entity, repository.



The view templates consist of the following:



Within the static package, the CSS templates crud.css and dashboard.css are present. Within the templates package, the HTML templates are present.

VI. Limitations

1. Misaligned Front-End Design – the design does not align with St. Paul University Manila
2. Back-End Validation – there is limited data validation in the controller class. Therefore, there are errors that are still not validated.
3. Lack of Service Class – there is no service class to separate CRUD methods within the controller class.
4. Hard Deletion – the project has yet to implement soft deletion, which is marking user records as inactive instead of deleting.
5. Inactive User View – queries all users even those who are inactive.
6. Misaligned Create Function – the create function should be aligned with user registration using facial recognition.

VII. Challenges Encountered

1. Limited Knowledge – members are hindered by lack of knowledge regarding the technology stack and dependencies. Specifically, the Spring framework and the frontend template engine Thymeleaf is not utilized by their full potential.
2. Limited Project and Time Management – there is no system development life cycle (SDLC). There is also a lack of knowledge in project management systems and versioning, specifically in using Git and GitHub.
3. Role Confusion – before the project there were no clear distinctions between each member within the project.